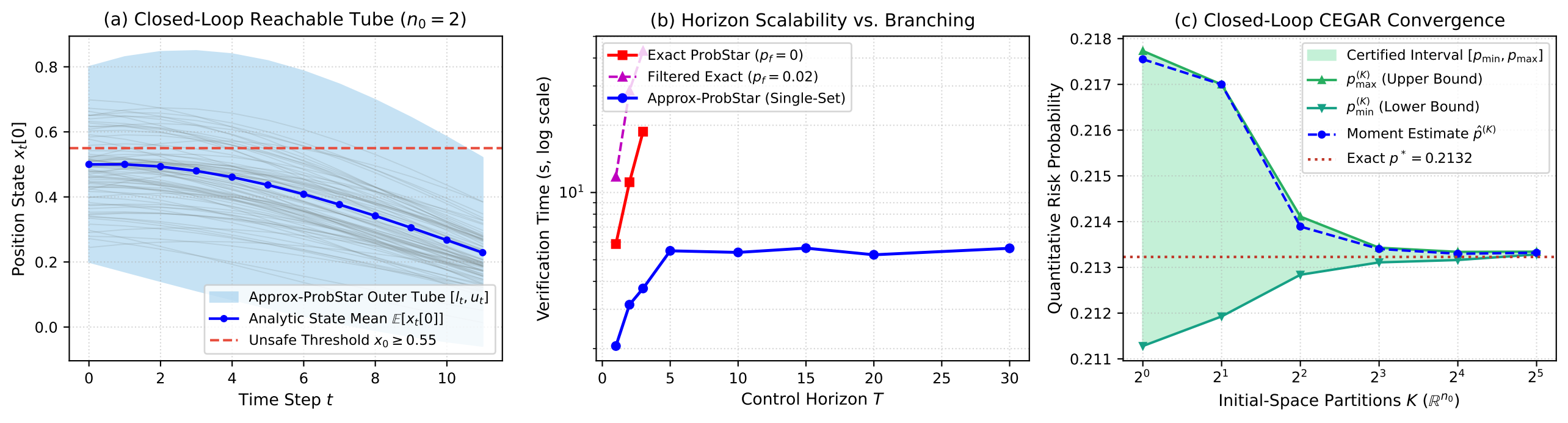
(a) Closed-Loop Reachable Tube ($n_0 = 2$)
Position State $x_t[0]$
Time Step t
Approx-ProbStar Outer Tube $[l_t, u_t]$
Analytic State Mean $\mathbb{E}[x_t[0]]$
Unsafe Threshold $x_0 \geq 0.55$
(b) Horizon Scalability vs. Branching
Verification Time (s, log scale)
Control Horizon T
Exact ProbStar ($p_f = 0$)
Filtered Exact ($p_f = 0.02$)
Approx-ProbStar (Single-Set)
(c) Closed-Loop CEGAR Convergence
Quantitative Risk Probability
Initial-Space Partitions K ($\mathbb{R}^{n_0}$)
Certified Interval $[p_{\min}, p_{\max}]$
$p_{\max}^{(K)}$ (Upper Bound)
$p_{\min}^{(K)}$ (Lower Bound)
Moment Estimate $\hat{p}^{(K)}$
Exact $p^* = 0.2132$